

Sobhan Soleymani

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PROFESSIONAL SUMMARY

Computer Vision and Machine Learning Scientist with 9+ years of experience in multimodal learning, computational photography, generative modeling, and edge AI systems. Proven track record of designing, optimizing, and deploying deep learning models for real-time perception, interaction, and image enhancement on resource-constrained devices in production environments. Expertise spans multimodal representation learning, generative models, transformers, biometrics, adversarial robustness, and hardware-aware optimization for XR/VR, wearable, and mobile platforms. Author of 40+ peer-reviewed papers with 1400+ citations in leading venues including NeurIPS, CVPR, ECCV, TBIOM, and WACV, and inventor on multiple patents in gesture recognition, multimodal interaction, and image restoration.

CORE EXPERTISE

Multimodal & Representation Learning: Multimodal fusion, quality-aware fusion, cross-modal learning, multimodal representation learning, semi-supervised learning, multi-sample multimodal learning

Computer Vision & Computational Photography: Image restoration, bokeh rendering, image dehazing, camera calibration, rectification, image synthesis

Generative Modeling: GANs, image-to-image translation, sketch-to-photo synthesis, visual synthesis

Robust & Trustworthy ML: Adversarial robustness, morph detection, cross-dataset generalization, biometric security

Edge AI & Real-Time ML Systems: Real-time inference, ONNX, quantization, hardware-aware optimization, resource-constrained deployment

Applications: XR/VR interaction, wearable AI, gesture recognition, biometrics, human-computer interaction

Tools & Frameworks: Python, C++, PyTorch, OpenCV, ONNX, TensorFlow, MATLAB

WORK EXPERIENCE

Senior Computer Vision & Computational Photography Research Scientist
Samsung R&D, San Diego, CA

June 2022–Present

- Led the development of real-time multimodal gesture recognition systems for XR/VR applications, enabling robust human-computer interaction across diverse deployment scenarios.
- Optimized and deployed deep learning models for real-time inference on edge devices under latency, memory, and compute constraints for wearable and mobile platforms.
- Designed saliency-aware and depth-guided bokeh rendering methods for computational photography applications.
- Built camera calibration and rectification pipelines for computational photography and XR/VR perception.
- Collaborated with cross-functional software and hardware teams to integrate ML systems into production pipelines under real-world device constraints.
- Led gesture-recognition data collection and annotation efforts, establishing scalable pipelines for dataset generation, training, benchmarking, and model evaluation.
- Converted computer vision models to ONNX and established testing and validation workflows for reliable on-device inference and deployment.
- Developed transformer-based and generative adversarial approaches for image dehazing and restoration under real-world degradation conditions.
- Designed dataset generation and evaluation frameworks for image dehazing across diverse visual conditions.

Postdoctoral Fellow, West Virginia University, Morgantown, WV

Sept. 2021–May 2022

- Led the development of biometric security pipelines for morph generation and detection using attention mechanisms, wavelet analysis, and disentangled representations.
- Designed multimodal biometric recognition frameworks using disentangled representations and attention-based fusion.
- Developed augmentation and semi-supervised learning methods to improve generalization and leverage unlabeled data.

- Developed unimodal and multimodal biometric recognition systems integrating face, speech, iris, and fingerprint modalities for single-sample and multi-sample identification.
- Designed methods to analyze and improve adversarial robustness in vision models, including adversarial example generation for face and iris recognition systems.
- Designed generative frameworks for unsupervised image-to-image translation and sketch-to-photo synthesis for style transfer and forensic applications.
- Developed representation learning and augmentation methods for semi-supervised and multimodal learning.
- Evaluated biometric recognition systems across datasets and real-world conditions to study robustness and generalization.
- Built preprocessing, feature extraction, dataset development, and evaluation pipelines to improve biometric recognition performance and robustness.

PATENTS

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| • Systems and Methods for Multi-Modal Interaction Analysis
Real-time interaction analysis, 3D hand pose estimation, gesture recognition | Granted, 2026 |
| • Modular Pipeline for High-Fidelity Hand-Arm Motion Synthesis and Multi-View Rendering
Hand-arm motion synthesis, synthetic data generation, multi-view rendering | Published, 2025 |
| • System and Method for Static and Dynamic Real-time Hand Gesture Recognition
Real-time gesture recognition, static-dynamic gesture fusion, XR interaction | Published, 2025 |
| • Generative Adversarial Networks for Transformer-Based Dehazing
Image dehazing, generative modeling, vision transformers | Published, 2025 |

SELECTED PUBLICATIONS (40+ PUBLICATIONS; 1400+ CITATIONS)

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| • Revisiting Outer Optimization in Adversarial Training | ECCV, 2022 |
| • Quality-Aware Multimodal Biometric Recognition | TBIOM, 2021 |
| • SuperMix: Supervising the Mixing Data Augmentation | CVPR, 2021 |
| • Self-Supervised Wasserstein Pseudo-Labeling for Semi-Supervised Image Classification | CVPR, 2021 |
| • Mutual Information Maximization on Disentangled Representations for Differential Morph Detection | WACV, 2021 |
| • Exploiting Joint Robustness to Adversarial Perturbations | CVPR, 2020 |
| • Transporting Labels via Hierarchical Optimal Transport for Semi-Supervised Learning | ECCV, 2020 |
| • SmoothFool: An Efficient Framework for Computing Smooth Adversarial Perturbations | WACV, 2020 |
| • Fast Geometrically-Perturbed Adversarial Faces | WACV, 2019 |
| • Unsupervised Image-to-Image Translation Using Domain-Specific Variational Information Bound | NeurIPS, 2018 |
| • Multi-Level Feature Abstraction from CNNs for Multimodal Biometric Identification | ICPR, 2018 |
| • Prosodic-Enhanced Siamese CNNs for Cross-Device Text-Independent Speaker Verification | BTAS, 2018 |

PROFESSIONAL SERVICE

Journal Reviewer: TPAMI, TNNLS, TIP, TIFS, TBIOM, IEEE JSTSP, IEEE Sensors Journal, SPL.

Conference Reviewer: CVPR, ICCV, ECCV, NeurIPS, ICLR, WACV, ICIP.

EDUCATION

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| Ph.D. in Electrical Engineering , West Virginia University, USA
Dissertation: <i>Deep Models for Improving the Performance and Reliability of Person Recognition</i> | May 2021 |
| M.Sc. in Electrical Engineering , École Polytechnique Fédérale de Lausanne (EPFL), Switzerland
Thesis: <i>Tomographic Field Reconstruction Using a Mobile Sensor Network</i> | March 2009 |
| B.Sc. in Electrical Engineering (Communications) , University of Tehran, Iran | July 2007 |